

Celebal Technologies Internship – Week 7 Assignment Delta Lake MERGE Implementation

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Internship: Celebal Technologies Data Engineering Internship

Week: 7

Objective

The objective of this assignment is to perform incremental data processing using **Delta Lake**. The assignment demonstrates how to load data into Delta tables, clean datasets by handling missing values and duplicate records, simulate incremental data, perform MERGE (UPSERT) operations, validate the final dataset, and understand how Delta Lake simplifies modern data engineering workflows.

Dataset Description

Two customer datasets were used in this assignment.

customer_master.csv

The master dataset contains customer information including:

- Customer ID
- First Name
- Last Name
- Email
- Phone Number
- City
- State
- Country
- Signup Date
- Membership
- Status

This dataset intentionally contains duplicate records and missing values to demonstrate data cleaning.

customer_incremental.csv

The incremental dataset contains:

- Existing customers with updated information
- New customer records

This dataset is merged into the master Delta table using Delta Lake MERGE.

Software Requirements

- Python
 - Apache Spark (PySpark)
 - Delta Lake
 - Google Colab (Execution Environment)
 - Pandas
-

Q1. Load Dataset into a Delta Table Explanation

The master customer dataset was loaded into a Spark DataFrame with automatic schema inference enabled. The DataFrame was then written as a Delta table using Delta Lake storage format.

Screenshot

Q1. Load Dataset into a Delta Table

```
from pyspark.sql.types import *

master_df = spark.read \
    .option("header", True) \
    .option("inferSchema", True) \
    .csv("customer_master.csv")

master_df.show(5)
```

customer_id	first_name	last_name	email	phone	city	state	country	signup_date	membership	status
CUST00001	Diana	Bass	paulerica@example...	9935730990	Mumbai	Delhi	India	2023-10-17	Platinum	Inactive
CUST00002	Patricia	Pena	samanthahoward@ex...	9489223748	Pune	Bihar	India	2022-09-08	Platinum	Active
CUST00003	Dalton	Smith	carolyn94@example...	(913)361-4471x19728	Jaipur	Maharashtra	India	2023-11-24	Platinum	Inactive
CUST00004	Rose	Hall	justinadkins@exam...	(329)594-8104x439	Delhi	Maharashtra	India	2022-02-02	Silver	Active
CUST00005	Phyllis	Cook	antonio04@example...	001-988-350-0848x...	Kolkata	Delhi	India	2023-06-02	Platinum	Active

only showing top 5 rows

```
master_df.printSchema()

root
 |-- customer_id: string (nullable = true)
 |-- first_name: string (nullable = true)
 |-- last_name: string (nullable = true)
 |-- email: string (nullable = true)
 |-- phone: string (nullable = true)
 |-- city: string (nullable = true)
 |-- state: string (nullable = true)
 |-- country: string (nullable = true)
 |-- signup_date: date (nullable = true)
 |-- membership: string (nullable = true)
 |-- status: string (nullable = true)
```

```
delta_master = spark.read \
    .format("delta") \
    .load("/content/customer_master_delta")

delta_master.show(5)
```

customer_id	first_name	last_name	email	phone	city	state	country	signup_date	membership	status
CUST00001	Diana	Bass	paulerica@example...	9935730990	Mumbai	Delhi	India	2023-10-17	Platinum	Inactive
CUST00002	Patricia	Pena	samanthahoward@ex...	9489223748	Pune	Bihar	India	2022-09-08	Platinum	Active
CUST00003	Dalton	Smith	carolyn94@example...	(913)361-4471x19728	Jaipur	Maharashtra	India	2023-11-24	Platinum	Inactive
CUST00004	Rose	Hall	justinadkins@exam...	(329)594-8104x439	Delhi	Maharashtra	India	2022-02-02	Silver	Active
CUST00005	Phyllis	Cook	antonio04@example...	001-988-350-0848x...	Kolkata	Delhi	India	2023-06-02	Platinum	Active

only showing top 5 rows

Q2. Handle Missing Values and Remove Duplicate Records Explanation

Missing values were identified using Spark SQL functions. Records containing null values were removed using `dropna()`, and duplicate records were removed using `dropDuplicates()`. The cleaned dataset was then written back to the Delta table.

Screenshot

Q2. Handle Missing Values and Remove Duplicate Records

```
from pyspark.sql.functions import col, count, when

master_df.select([
    count(when(col(c).isNull(), c)).alias(c)
    for c in master_df.columns
]).show()
```

customer_id	first_name	last_name	email	phone	city	state	country	signup_date	membership	status
0	0	0	0	80	0	0	0	0	50	0

```
clean_df = master_df.dropna()
```

```
clean_df = clean_df.dropDuplicates()
```

```
print("Original Rows:", master_df.count())
print("Clean Rows:", clean_df.count())
```

```
Original Rows: 5030
Clean Rows: 4900
```

```
clean_df.write \
    .format("delta") \
    .mode("overwrite") \
    .save("/content/customer_master_delta")
```

Q3. Load Incremental Dataset Explanation

The incremental dataset containing updated and new customer records was loaded into Spark. Duplicate customer IDs were removed to ensure that each customer record participated only once in the MERGE operation.

Screenshot

```
Q3. Load Incremental Dataset

incremental_df = spark.read \
    .option("header", True) \
    .option("inferSchema", True) \
    .csv("customer_incremental.csv")

incremental_df.show(5)
```

customer_id	first_name	last_name	email	phone	city	state	country	signup_date	membership	status
CUST02526	Allison	Jones	torresryan@exampl...	+1-935-467-4849x6...	Delhi	Bihar	India	2023-08-20	Platinum	Active
CUST02923	James	Martin	cooperamanda@exam...	887.364.6634x853	Delhi	Bihar	India	2024-01-11	Silver	Inactive
CUST00890	Robert	Wade	david74@example.net	001-886-900-1466	Patna	Telangana	India	2024-04-16	Platinum	Active
CUST00658	Darlene	Holland	beckdiane@example...	+1-648-500-1309	Hyderabad	Telangana	India	2023-02-26	Platinum	Active
CUST04317	Jeffrey	Mitchell	jessica48@example...	250.992.1465x6318	Patna	Maharashtra	India	2024-04-28	Silver	Inactive

```
only showing top 5 rows

print("Incremental Rows:", incremental_df.count())

Incremental Rows: 500

incremental_df = incremental_df.dropDuplicates(["customer_id"])

print("Rows After Removing Duplicates:", incremental_df.count())

Rows After Removing Duplicates: 499
```

Q4. Apply MERGE Operation Explanation

The Delta Lake MERGE (UPSERT) operation synchronized the incremental dataset with the master Delta table by updating existing customer records and inserting new records that did not already exist. The MERGE operation performed:

- Updating existing customer records.
- Inserting new customer records.
- Maintaining a single consistent customer record for every customer ID.

Screenshot

```
Q4. Apply MERGE Operation (Update Existing and Insert New Records)

23] from delta.tables import DeltaTable
    delta_table = DeltaTable.forPath(
        spark,
        "/content/customer_master_delta"
    )

24] (
    delta_table.alias("target")
    .merge(
        incremental_df.alias("source"),
        "target.customer_id = source.customer_id"
    )
    .whenMatchedUpdate(
        set={
            "first_name": "source.first_name",
            "last_name": "source.last_name",
            "email": "source.email",
            "phone": "source.phone",
            "city": "source.city",
            "state": "source.state",
            "country": "source.country",
            "signup_date": "source.signup_date",
            "membership": "source.membership",
            "status": "source.status"
        }
    )
    .whenNotMatchedInsert(
        values={
            "customer_id": "source.customer_id",
            "first_name": "source.first_name",
            "last_name": "source.last_name",
            "email": "source.email",
            "phone": "source.phone",
            "city": "source.city",
            "state": "source.state",
            "country": "source.country",
            "signup_date": "source.signup_date",
            "membership": "source.membership",
            "status": "source.status"
        }
    )
    .execute()
)

... DataFrame[num_affected_rows: bigint, num_updated_rows: bigint, num_deleted_rows: bigint, num_inserted_rows: bigint]
```

Q5. Validate Results

Explanation

The final Delta table was validated by checking:

- Final row count
- Duplicate customer IDs
- Dataset summary

Validation confirmed that the MERGE operation completed successfully.

Output Summary

Dataset	Rows
Original Dataset	5030
Clean Dataset	4900
Incremental Dataset	499
Final Dataset	5119

Duplicate Customer IDs 0

The validation confirms that the MERGE operation successfully updated the Delta table while preserving data integrity.

Screenshot

Q5. Validate Results

```
final_df = spark.read \
    .format("delta") \
    .load("/content/customer_master_delta")
```

```
print("Final Row Count:", final_df.count())
```

```
... Final Row Count: 5119
```

```
from pyspark.sql.functions import col

duplicate_count = (
    final_df.groupBy("customer_id")
    .count()
    .filter(col("count") > 1)
    .count()
)

print("Duplicate Customer IDs:", duplicate_count)

Duplicate Customer IDs: 0
```

Q6. Display Final Dataset and Summary

```
final_df.show(20, truncate=False)
```

```
*** +-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|customer_id|first_name|last_name|email|phone|city|state|country|signup_date|membership|status|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|CUST00001|Diana|Bass|nauferica@example.org|9935730990|Mumbai|Delhi|India|2023-10-17|Platinum|Inactive|
|CUST00002|Patricia|Pena|samanthahoward@example.net|9489223748|Pune|Bihar|India|2022-09-08|Platinum|Active|
|CUST00003|Dalton|Smith|carolyn94@example.com|(913)361-4471x19728|Jaipur|Maharashtra|India|2023-11-24|Platinum|Inactive|
|CUST00004|Rose|Hall|justinadkins@example.com|(329)594-8104x430|Delhi|Maharashtra|India|2022-02-02|Silver|Active|
|CUST00005|Phyllis|Cook|antonio04@example.net|001-988-350-0848x43155|Kolkata|Delhi|India|2023-06-02|Platinum|Active|
|CUST00006|Wesley|Martin|emoore@example.com|859-552-2841x0783|Jaipur|Uttar Pradesh|India|2024-01-04|Silver|Inactive|
|CUST00007|Daniel|Campbell|tterrell@example.net|(579)822-1239x4281|Hyderabad|Delhi|India|2023-08-27|Silver|Inactive|
|CUST00008|Brandy|Brewer|vanderson@example.net|+1-343-662-7124x976|Hyderabad|Bihar|India|2022-12-15|Silver|Inactive|
|CUST00009|Edward|Love|bishopeugene@example.org|737-303-1192x667|Delhi|Delhi|India|2022-04-15|Silver|Active|
|CUST00010|Hunter|Johnson|andrewyoung@example.com|836-828-0929x38037|Chennai|Telangana|India|2024-05-17|Silver|Inactive|
|CUST00011|William|Johnson|john80@example.com|+1-664-309-1475x815|Mumbai|Uttar Pradesh|India|2023-07-04|Silver|Inactive|
|CUST00012|Sheila|Martin|melissa18@example.net|517-821-0574|Kolkata|Karnataka|India|2024-04-29|Platinum|Active|
|CUST00013|Rebecca|Hill|wrightsarah@example.net|5273704592|Mumbai|Delhi|India|2023-12-23|Platinum|Active|
|CUST00014|Richard|Davis|christopherboyd@example.net|824-298-4604x50411|Hyderabad|Maharashtra|India|2024-03-02|Silver|Active|
|CUST00015|Jennifer|Jimenez|susan80@example.net|998-728-6546|Hyderabad|Rajasthan|India|2023-01-25|Platinum|Inactive|
|CUST00016|Jonathan|Harper|joellewis@example.org|459-739-7629x137|Chennai|Karnataka|India|2022-06-16|Silver|Inactive|
|CUST00017|Sonia|Young|joe6@example.com|218-287-8749|Mumbai|Bihar|India|2023-12-20|Platinum|Active|
|CUST00018|Kathleen|Mitchell|kennethclark@example.org|(525)298-4346|Lucknow|Uttar Pradesh|India|2022-06-17|Gold|Active|
|CUST00019|April|Frederick|julianadams@example.com|(939)663-7638x01698|Chennai|Delhi|India|2023-12-03|Silver|Active|
|CUST00020|Eileen|Cunningham|malone.linda@example.net|784-963-1883x364|Chennai|Uttar Pradesh|India|2024-04-04|Gold|Active|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
only showing top 20 rows
```

```
final_df.printSchema()
```

```
root
|-- customer_id: string (nullable = true)
|-- first_name: string (nullable = true)
|-- last_name: string (nullable = true)
|-- email: string (nullable = true)
|-- phone: string (nullable = true)
|-- city: string (nullable = true)
|-- state: string (nullable = true)
|-- country: string (nullable = true)
|-- signup_date: date (nullable = true)
|-- membership: string (nullable = true)
|-- status: string (nullable = true)
```

```
print("Original Dataset :", master_df.count())
print("Clean Dataset :", clean_df.count())
print("Incremental Dataset :", incremental_df.count())
print("Final Dataset :", final_df.count())
```

```
Original Dataset : 5030
Clean Dataset : 4900
Incremental Dataset : 499
Final Dataset : 5119
```


Conclusion

This assignment successfully demonstrated incremental data processing using Delta Lake. The master customer dataset was cleaned by removing missing values and duplicate records before being stored as a Delta table. An incremental dataset containing updated and new customer records was then merged into the Delta table using the MERGE operation. Finally, the merged dataset was validated to ensure correct row counts and the absence of duplicate customer IDs. This assignment provided practical experience with Delta Lake's ACID transactions, efficient UPSERT operations, and incremental data processing techniques commonly used in modern data engineering pipelines.

References

- [Apache Spark Documentation](#)
- [Delta Lake Documentation](#)
- [PySpark API Documentation](#)
- [Celebal Technologies Data Engineering Internship Material](#)